Lab 3 — Applying requirements engineering to FoodGO

Course: CS 488 AI-First Software Engineering, Fall 2026 · **Team:** Hamza Ben Labiedh, Ali Chamakh, Mohamed Krid · **Date:** 1 Nov 2026 **PDF of this worksheet:** `LAB3_REQUIREMENTS_WORKSHEET.pdf` · **SRS skeleton:** `SRS_SKELETON.md` · **Full specification (our earlier Lab 3 deliverable):** `../lab3-specification/FoodGO_Software_Specification.pdf`

Project pitch. FoodGO lets employees in **Lac 3 / Zone Industrielle Le Kram** pre-order lunch in the morning, so they stay at their desks at noon and the area's lunchtime traffic drops. The restaurant's own delivery staff bring the food.

How this worksheet was built. Our team already turned its validated elicitation document into a full specification (FR-01–FR-30, BR-01–BR-16, quality requirements, 36 unresolved items). This worksheet walks the six Lab 3 steps over that material and adds what changed since: FR-31 (AI forecast, ADR-0004) and FR-32 / BR-17 (service area, ADR-0007).

> **Honest notes.** Prepared by Hamza with Claude; Ali and Mohamed review it at the team meeting on 2 Nov. The **role-play interviews of Step 1 have not been run yet**: the guide below is ready for them, and their findings rows are empty on purpose. Step 5's review comes from an **independent AI reviewer** because no other team was available; a peer team's review is still to add. Instructor checkpoints are **not** signed yet.

Step 1 — Elicit requirements

Stakeholder list

Stakeholder	Interest / concern
Customer — employee or worker in Lac 3 / ZI Le Kram	Order lunch quickly in the morning; get it on time at noon without driving out; see where the order is; pay cash or online
Customer — student at an institution in the area	Same as employees; spec keeps students as a customer type, the area decides who is served
Restaurant owner	Predictable demand, fewer noon rushes, control over who may order (BR-04)
Restaurant manager	Set the ordering deadline and payment options; register customers in person
Restaurant staff	See the day's orders early, prepare exact quantities, assign their own couriers
Delivery person (employed by the restaurant)	See only their own deliveries with the address, phone, items and cash to collect

Stakeholder	Interest / concern
Platform administrator	Provision restaurants and staff accounts; monitor activity without touching orders
Platform owner	A product that works and can grow; reputation; no delivery workforce of its own (BR-06)
Payment provider (Tunisian e-payment)	A clean integration; correct payment status
Workplace / campus organization in the area	Employees eat on time and come back on time; fewer lunchtime trips around the site
SMS / notification service	Clear contract for sending one-time codes and notices

Elicitation log

Technique	Participant	Key findings (raw notes)
Document analysis	Our validated Requirements Elicitation Document (Lab 2), summarised in the specification	Lunch is ordered by everyone in the same short window; restaurants are swamped and cook without knowing demand. Customers must be registered **in person** by the restaurant, no self-sign-up. Restaurant delivers with its own staff and equipment. Cash on delivery plus Tunisian e-payment. Customers want to see order progress. Many failure cases (cancel, refund, failed delivery) left open.
Product-owner interview (live, 1 Nov 2026)	Hamza, acting as product owner	"The app idea is not simply preordering food … only in **Lac 3, zone industrielle Kram**, to avoid **heavy traffic during lunchtime** in the area." Follow-up: "Lac 3, zone indus Kram are the **same zone**." Follow-up: "**Outside the area cannot join** FoodGO." → FR-32, BR-17, ADR-0007; also settles the spec's unresolved item 35 (initial deployment area).
Prototype walkthrough	Team, on the midterm MVP	The deadline must be visible before ordering; staff want the order queue live; couriers need the cash amount up front. (Used for NFR-07 and the UI.)
Role-play interview 1	*Interviewer: ____ · Persona: an employee at a Lac 3 office*	*to fill in class (2 Nov)*
Role-play interview 2	*Interviewer: ____ · Persona: staff at a restaurant in ZI Le Kram*	*to fill in class (2 Nov)*

Role-play interview guide (10 minutes each).

- *Employee persona:* "Walk me through your lunch on a normal workday, from the moment you get hungry." · "How long does it take you to get food and come back? What slows you down?" · "What would make you order the evening before or in the morning instead?" · "How would you want to pay, and how do you want to know your food is coming?"

- *Restaurant persona:* "What does your kitchen look like between 11:30 and 13:30?" · "How do you decide how much to cook?" · "Who delivers today, and how do you organise them?" · "When would you need the last order to be able to cook everything on time?"

☐ Checkpoint: show our stakeholder list to the instructor — *to do in class*

Step 2 — Analyze & classify

ID	Type	Category	Description (from the raw notes)	Priority	Source
C-01	FR	Access	Restaurant staff register a customer in person; no self-sign-up	Must	Elicitation doc → FR-01–FR-04, BR-16
C-02	FR	Access	Customer signs in with phone number and a one-time code	Must	Elicitation doc → FR-03, BR-05
C-03	FR	Service area	Serve only restaurants and workplaces in Lac 3 / ZI Le Kram; restaurants outside cannot join	Must	PO interview → FR-32, BR-17
C-04	FR	Ordering	Customer sees only restaurants that registered them, with menus and availability	Must	Elicitation doc → FR-05–FR-08
C-05	FR	Ordering	Order before the restaurant's deadline; refused after it	Must	Elicitation doc → FR-12, FR-13, BR-01
C-06	FR	Ordering	Exact total and a unique order number	Must	Elicitation doc → FR-11, FR-14
C-07	FR	Delivery	Restaurant assigns its own delivery person before dispatch	Must	Elicitation doc → FR-19, BR-06, BR-10
C-08	FR	Delivery	Courier sees only their assignments and confirms delivery	Must	Elicitation doc → FR-20–FR-23, BR-08, BR-11
C-09	FR	Payment	Cash on delivery or Tunisian e-payment; payment status recorded	Must (cash) / Should (real e-payment provider)	Elicitation doc → FR-25–FR-28
C-10	FR	Tracking	Customer follows order status live	Should	Elicitation doc → FR-24
C-11	FR	Planning	Forecast of tomorrow's dishes, advice only	Should	Team decision → FR-31, ADR-0004
C-12	NFR	Performance	Pages and API feel instant; ordering under 2 s	Must	Spec §7 → NFR-01

ID	Type	Category	Description (from the raw notes)	Priority	Source
C-13	NFR	Security	Roles, restaurant scope, hashed credentials, rate limits	Must	Spec §7, §12 → NFR-02
C-14	NFR	Privacy	Customer data only for authorised parties; none sent to the LLM	Must	Spec §7, BR-15 → NFR-03
C-15	NFR	Reliability	No lost, duplicate or half-written orders	Must	Spec §7 → NFR-04
C-16	NFR	Usability	Ordering in a few screens; deadline visible	Should	Prototype walkthrough → NFR-07
C-17	NFR	Accessibility	WCAG 2.2 AA	Should	Spec §7 → NFR-08
C-18	FR	Order lifecycle	Customer cancels an order	Won't (this release)	Elicitation doc: unresolved (cancellation)
C-19	FR	Delivery	Live GPS tracking of the courier	Won't	Spec §20.11 do-not-implement list

☐ Checkpoint: instructor reviews at least three requirements for clarity — *to do in class*

Step 3 — Final requirement statements (draft, before peer review)

Functional requirements keep the identifiers of our specification (FR-01–FR-30) and add FR-31 and FR-32. Business rules BR-01–BR-17 constrain them (BR-17 is new). Non-functional requirements are NFR-01–NFR-12 from our backlog.

ID	Requirement statement	Priority
FR-01	The system shall support customer account creation.	Must
FR-02	The system shall collect the customer's full name.	Must
FR-03	The system shall collect the customer's phone number.	Must
FR-04	The system shall collect the customer's workplace or educational institution.	Must
FR-05	The system shall display available restaurants to the customer.	Must
FR-06	The system shall allow customers to view restaurant menus.	Must
FR-07	The system shall allow restaurants to manage their menus.	Must
FR-08	The system shall allow restaurants to mark menu items available or unavailable.	Must
FR-09	The system shall allow customers to select menu items.	Must

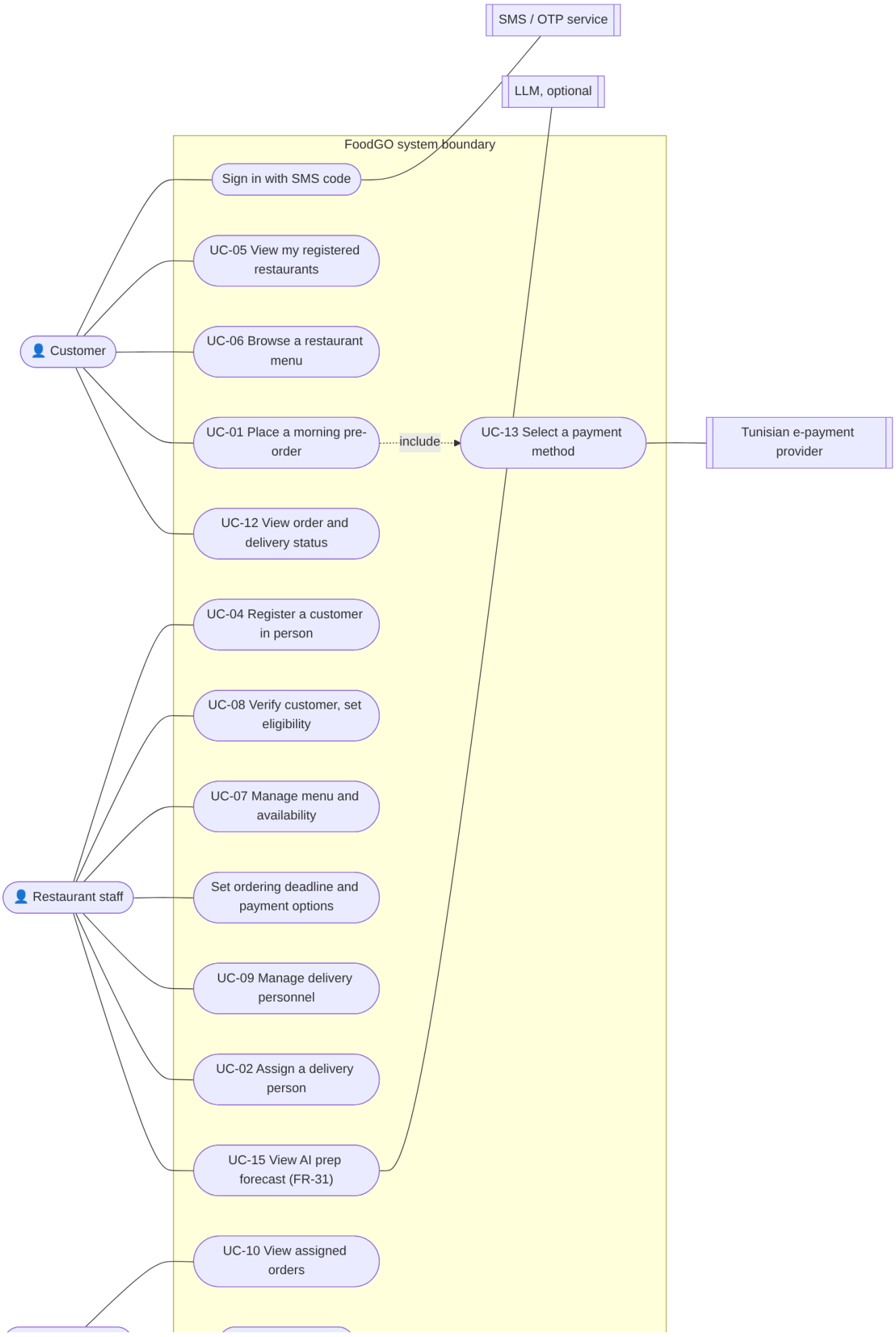
ID	Requirement statement	Priority
FR-10	The system shall allow customers to specify quantities.	Must
FR-11	The system shall calculate the order total from item prices and quantities.	Must
FR-12	The system shall accept pre-orders before the restaurant-defined ordering deadline.	Must
FR-13	The system shall prevent orders after the applicable ordering deadline.	Must
FR-14	The system shall confirm submitted orders with a unique identifier.	Must
FR-15	The system shall make the submitted order available to the corresponding restaurant.	Must
FR-16	The system shall provide authorized restaurant personnel with the customer information needed for verification.	Must
FR-17	The system shall support restaurant verification that a customer belongs to the applicable target organization.	Must
FR-18	The system shall allow restaurants to manage their delivery personnel.	Must
FR-19	The system shall allow restaurants to assign orders to delivery personnel.	Must
FR-20	The system shall allow delivery personnel to view their assigned orders.	Must
FR-21	The system shall provide delivery personnel with the information necessary to perform assigned deliveries.	Must
FR-22	The system shall allow delivery personnel to update delivery status.	Must
FR-23	The system shall allow delivery personnel to confirm successful delivery.	Must
FR-24	The system shall allow customers to view order and delivery status.	Should
FR-25	The system shall allow customers to select an available payment method.	Must
FR-26	The system shall support payment upon delivery where enabled by the restaurant.	Must
FR-27	The system shall support approved third-party Tunisian electronic payment services where integrated.	Should
FR-28	The system shall record payment status against the order.	Must
FR-29	The system shall allow the platform to provision accounts for restaurant staff.	Must
FR-30	The system shall allow restaurants to provision accounts for their delivery personnel.	Must
FR-31	The system shall give restaurant staff an advisory forecast of the next service's dishes, which never changes orders, payments, eligibility or the menu.	Should
FR-32	The system shall record a service zone for every restaurant and workplace and accept only Lac 3 / Zone Industrielle Le Kram.	Must
BR-17	An order shall be accepted only when both the restaurant and the customer's workplace are inside the service area.	Must
NFR-01	95 % of API requests shall complete in under 500 ms and 99 % in under 1.5 s; order submission in under 2 s.	Must

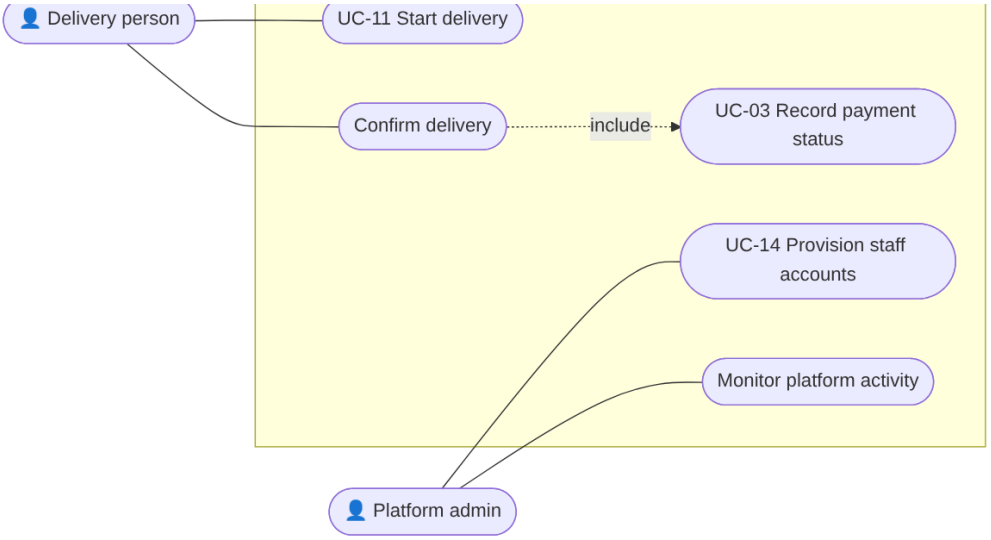
ID	Requirement statement	Priority
NFR-02	The system shall authenticate users, enforce role and restaurant-scoped authorisation, hash credentials and rate-limit sign-in.	Must
NFR-03	Customer data shall be visible only to authorised parties, and no personal data shall be sent to the LLM.	Must
NFR-04	The system shall not lose, duplicate or half-write an order.	Must
NFR-05	Business rules shall be separated from the UI and persistence; complexity at most 10 per function.	Should
NFR-06	Each feature shall have unit and integration tests; line coverage at least 60 %.	Must
NFR-07	A customer shall be able to place an order in at most 4 screens, with the deadline visible before ordering.	Should
NFR-08	The user interface shall meet WCAG 2.2 AA.	Should
NFR-09	Security-relevant actions shall be written to an audit log; a health endpoint shall exist.	Should
NFR-10	Payment, SMS/OTP, notification and LLM providers shall be reached through replaceable adapters.	Should
NFR-11	The system shall be available 99.5 % of the time during service hours.	Should
NFR-12	The system shall run with Node.js 22+ and no external service.	Could

☐ Checkpoint: read our top three Must requirements aloud to the instructor (FR-13, FR-32, BR-17) — *to do in class*

Step 4 — Use cases

Actors: Customer, Restaurant staff, Delivery person, Platform administrator. External systems: Tunisian e-payment provider, SMS/OTP service, LLM (optional).


SMS / OTP service
LLM, optional
FoodGO system boundary
Sign in with SMS code
UC-05 View my registered restaurants
Customer
UC-06 Browse a restaurant menu
UC-01 Place a morning pre-order
include
UC-13 Select a payment method
Tunisian e-payment provider
UC-12 View order and delivery status
UC-04 Register a customer in person
UC-08 Verify customer, set eligibility
UC-07 Manage menu and availability
Restaurant staff
Set ordering deadline and payment options
UC-09 Manage delivery personnel
UC-02 Assign a delivery person
UC-15 View AI prep forecast (FR-31)
UC-10 View assigned orders



Use case UC-01 — Place a morning pre-order

- **Use Case ID:** UC-01
- **Name:** Place a morning pre-order
- **Actor:** Customer (an employee registered by the restaurant)
- **Preconditions:** The customer is signed in; a restaurant registered them and they are eligible; the restaurant and the customer's workplace are in Lac 3 / Zone Industrielle Le Kram; it is before the restaurant's ordering deadline.
- **Main flow:**
    1. The customer opens one of their restaurants and sees today's menu and the time left before the deadline.
    2. The customer chooses available dishes and quantities; the system shows the exact total.
    3. The customer chooses cash on delivery or e-payment (only methods the restaurant enabled).
    4. The customer submits the order.
    5. The system checks registration, service area, deadline, availability and payment method, saves the order, and shows an order number FG-XXXXXX.
    6. The restaurant sees the new order in its queue immediately.
- **Alternate flow:** 5a. The deadline has passed → the order is refused ("Ordering for today closed at HH:MM", ORDER-001). 5b. A dish became unavailable → refused (ORDER-002). 5c. The restaurant or workplace is outside the service area → refused (ORDER-003). 5d. E-payment is declined → no order is created (PAY-001). 4a. The customer's connection drops and the app retries → the same order is returned, not a second one.
- **Postconditions:** One order with its items, total, payment record and deadline snapshot exists, visible to this customer and this restaurant only.
- **Linked requirements:** FR-05, FR-06, FR-09–FR-15, FR-25, FR-26, FR-32; BR-01, BR-02, BR-04, BR-12, BR-16, BR-17; NFR-01, NFR-04

Use case UC-04 — Register a customer in person

- **Use Case ID:** UC-04
- **Name:** Register a customer in person
- **Actor:** Restaurant staff

- **Preconditions:** The staff member is signed in to their own restaurant; the customer is present with an ID.
- **Main flow:**
    1. The staff member opens Customers and enters the customer's full name and phone number.
    2. The staff member picks the customer's workplace (each workplace is in the service area).
    3. The staff member checks the customer's identity and ticks "I checked their identity in person".
    4. The system creates the customer (or reuses the existing person with that phone number) and adds a membership for this restaurant.
    5. The customer can now sign in with an SMS code and sees this restaurant.
- **Alternate flow:** 2a. The workplace is missing → the staff member adds it with its zone; a place outside the area is refused (VAL-001). 3a. Identity not confirmed → registration refused (BR-05). 4a. The customer is already registered at this restaurant → refused as a duplicate.
- **Postconditions:** One canonical customer record and one active, eligible membership for this restaurant; nothing visible to other restaurants.
- **Linked requirements:** FR-01–FR-04, FR-16, FR-17, FR-32; BR-04, BR-05, BR-15, BR-16

☐ Checkpoint: show our use-case diagram to the instructor — *to do in class*

Step 5 — Peer review & validation

Reviewing team. No other team was available for the swap, so an **independent AI reviewer** (a fresh Claude agent that knew nothing about the project) applied the lab's validation checklist to our specification, backlog and the two ADRs that added requirements. It changed nothing. We checked every issue against the documents before accepting it. A peer team's review is still to add in class.

Validation checklist (applied to our own list by the reviewer): ☑ no contradictions — 3 found and fixed · ☑ no duplicates — 1 found and merged · ☑ every requirement testable — vague verbs and number-free NFRs fixed · ☑ no implementation details — tool names and code identifiers removed · ☑ consistent terminology — glossary added.

Reviewing team	Issue found	Our fix
Independent AI reviewer	**FR-31 contradicts the specification**, which puts AI out of scope (§2.2, AI-001, BG-03, do-not-implement item 05), yet the backlog shows it done.	FR-31 reworded to say exactly what it may and may not touch, and kept **Proposed**. SRS §2 records that the team withdrew A-13 (ADR-0004) and that stakeholder validation is pending.
Independent AI reviewer	**FR-32/BR-17 narrow the scope, but the spec still promises "many campuses and zones"**; items 10 and 35 still say unresolved; the joining rule had no number; FR-32 named code (`LAC3_ZI_KRAM`, migration).	Items 35 and 10 marked resolved by ADR-0007; joining rule numbered **BR-18** (also in the code, tests and traceability); FR-32 rewritten without code; new **NFR-13 Scalability** for the area's peak, target set after Sprint 2 interviews.

Reviewing team	Issue found	Our fix
Independent AI reviewer	**FR-01 contradicts BR-16:** "support customer account creation" reads like self-sign-up, which BR-16 and §20.11 forbid.	FR-01 rewritten: staff create the account for their own restaurant; no self-registration.
Independent AI reviewer	**Vague, untestable verbs:** "manage" (FR-07, FR-18), "support" (FR-17, FR-26, FR-27); acceptance criteria that repeat the requirement.	Each rewritten with observable behaviour: FR-07 create/edit/remove items; FR-17 ineligible customer can neither see nor order; FR-26 pending → paid when cash collected; FR-27 order not created if payment declined; FR-11 exact to the millime. FR-18 limited to create and list (deactivation stays unresolved).
Independent AI reviewer	**Deadline stated three times (FR-12, FR-13, BR-01) and the boundary undefined.**	FR-12 "strictly before", FR-13 "at or after → rejected, deadline shown". Identifiers kept for traceability.
Independent AI reviewer	**NFRs without numbers or with tool names:** no load for NFR-01; NFR-11's service window undecided; helmet, express-rate-limit, Node.js 22, bcrypt named.	NFR-01 states its load (300 orders at one restaurant); NFR-11 flagged untestable until the service window is decided; NFR-02, 05, 10, 12 rewritten as qualities, tools moved to the SDD.
Independent AI reviewer	**Inconsistent terms:** workplace / organization / institution / campus; account / access / credentials; "courier"; area names.	One glossary term each in SRS §1 (Customer account, Workplace, Service area, Delivery person), used throughout Sections 4–5.
Peer team (in class)	*to fill in*	*to fill in*

What the reviewer said we do well: every story traces to FR/BR identifiers, unresolved items are marked blocked rather than guessed, and the Validated / Proposed / Unresolved labels grade each item honestly.

Step 6 — Draft SRS skeleton

Sections 1–7 with Sections 4 and 5 completed (after the Step 5 revisions): `SRS_SKELETON.md`.

End-of-session submission checklist

- ☑ Stakeholder list for our project
- ☑ Elicitation log (document analysis, product-owner interview, prototype walkthrough; role-play interviews still to run)
- ☑ Requirements analysis table (19 candidates, classified and prioritised)
- ☑ Final requirements list with unique IDs

- ☑ Use-case diagram + two detailed use-case descriptions
- ☑ Peer review feedback log and revisions made (independent AI reviewer; a peer team's review still to add)
- ☑ Draft SRS skeleton with Sections 4–5 completed
- ☐ Instructor sign-off at each checkpoint — *not yet obtained*

FoodGO — Software Requirements Specification (draft skeleton)

Lab 3, Step 6 deliverable. Structure from Lab 2 §6 (IEEE 830 / ISO/IEC/IEEE 29148), Sections 1–7. As the lab asks, **Sections 4 and 5 are complete**: they are our Step 3 list after the Step 5 peer-review revisions. The other sections are short placeholders that point to the full specification (`../lab3-specification/FoodGO_Software_Specification.pdf`).

Version: 0.2 draft, 1 Nov 2026 · **Team:** Hamza Ben Labiedh, Ali Chamakh, Mohamed Krid

Labels. *Validated* = from our validated elicitation document. *Team decision* = decided by the team after Lab 3, recorded in an ADR. *Proposed* = a position we took to make the system buildable; still to be validated with stakeholders.

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1. Introduction *(placeholder)*

- **Purpose:** the requirements of FoodGO, for the team, the instructor and future maintainers.
- **Scope:** pre-order lunch for workplaces in **Lac 3 / Zone Industrielle Le Kram** (one area, two names), to cut lunchtime traffic there. The restaurant prepares and delivers with its own staff; the platform coordinates.
- **Definitions** (one term each, used everywhere below):
  - **Customer** — an employee or student whose workplace is in the service area.
  - **Customer account** — the access a restaurant creates for a customer in person; one account per person, one membership per restaurant.
  - **Workplace** — the office, factory, school or institution a customer belongs to (stored as an *organization*); it is always inside the service area.
  - **Service area** — Lac 3 / Zone Industrielle Le Kram.
  - **Delivery person** — a person employed and equipped by a restaurant to deliver its orders (not "courier").
  - **Ordering deadline** — the restaurant's daily cut-off time for the same day's orders.
- **References:** Lab 3 specification; Lab 4 SDD; ADR-0004 (AI forecast); ADR-0007 (service area); backlog.

2. Overall description *(placeholder)*

- **Product perspective:** a web application with four interfaces (customer, restaurant, delivery person, administrator) and adapters to an e-payment provider, an SMS/OTP service and an optional LLM.
- **User classes:** see the stakeholder list in the Lab 3 worksheet.
- **Constraints:** BR-01–BR-18; the do-not-implement list (spec §20.11) still applies except item 05, AI prediction, which the team withdrew for advisory use only (ADR-0004, Proposed).
- **Assumptions:** spec A-01–A-15, except A-13 (AI out of scope), withdrawn by the team on 1 Nov 2026.

- **Resolved since the specification:** unresolved item 35 *Initial deployment area* → Lac 3 / Zone Industrielle Le Kram; item 10 *Delivery-area rules* → partly (one area; restaurants outside it cannot join). Both by ADR-0007.

3. External interface requirements *(placeholder)*

User interfaces per actor (spec §13); REST JSON API documented in `docs/api/openapi.yaml`; payment, SMS/OTP, notification and LLM providers behind adapters (provider choice still unresolved).

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4. Functional requirements

Identifiers are kept from the specification so traceability does not break. Wording changed after the peer review is marked **(rev.)**.

4.1 Accounts and access

ID	Requirement	Priority	Status
FR-01	**(rev.)** The system shall allow authorized restaurant staff to create a customer account, valid only for their own restaurant. The system shall not offer customer self-registration.	Must	Validated
FR-02	The system shall store the customer's full name and show it only to authorized parties.	Must	Validated
FR-03	**(rev.)** The system shall store the customer's phone number and reject a number that is not in the form +216 followed by 8 digits.	Must	Validated
FR-04	**(rev.)** The system shall store the customer's workplace.	Must	Validated
FR-29	The system shall allow the platform administrator to create restaurant staff accounts, each bound to exactly one restaurant.	Must	Validated
FR-30	The system shall allow a restaurant to create accounts for its own delivery persons, visible to no other restaurant.	Must	Validated

4.2 Service area

ID	Requirement	Priority	Status
FR-32	**(rev.)** The system shall record, for every restaurant and every workplace, that it is located in the service area, and shall refuse to create a restaurant or workplace located anywhere else.	Must	Team decision (ADR-0007)

4.3 Restaurants and menus

ID	Requirement	Priority	Status
FR-05	The system shall show a customer only the restaurants that created their account and still consider them eligible.	Must	Validated
FR-06	The system shall show a customer the menu of each of those restaurants, including dishes that are currently unavailable, marked as such.	Must	Validated
FR-07	**(rev.)** The system shall allow authorized staff of a restaurant to create, edit and remove that restaurant's menu items (name, description, price).	Must	Validated
FR-08	The system shall allow authorized restaurant staff to mark a menu item available or unavailable; an unavailable item cannot be ordered.	Must	Validated

4.4 Ordering

ID	Requirement	Priority	Status
FR-09	The system shall allow a customer to add available menu items to an order.	Must	Validated
FR-10	The system shall allow a customer to set a quantity from 1 to 50 for each item.	Must	Validated
FR-11	**(rev.)** The system shall calculate the order total as the sum of quantity × unit price over all items, exact to the millime.	Must	Validated
FR-12	**(rev.)** The system shall accept an order only if it is submitted strictly before its restaurant's ordering deadline for that day.	Must	Validated
FR-13	**(rev.)** The system shall reject an order submitted at or after the deadline and tell the customer the deadline time.	Must	Validated
FR-14	The system shall confirm each accepted order with a unique identifier, and a retried submission shall return the same order rather than create a second one.	Must	Validated
FR-15	The system shall make an accepted order visible to its restaurant, and to no other restaurant, as soon as it is accepted.	Must	Validated

4.5 Customer verification

ID	Requirement	Priority	Status
FR-16	The system shall show authorized staff of a restaurant the name, phone number and workplace of the customers registered at that restaurant.	Must	Validated
FR-17	**(rev.)** The system shall allow authorized restaurant staff to mark a customer ineligible; an ineligible customer shall neither see nor order from that restaurant.	Must	Validated

4.6 Delivery

ID	Requirement	Priority	Status
FR-18	**(rev.)** The system shall allow a restaurant to create and list its own delivery persons. Deactivation and other management details remain unresolved (spec item 12).	Must	Validated
FR-19	The system shall allow authorized restaurant staff to assign an order to one of that restaurant's active delivery persons.	Must	Validated
FR-20	The system shall show a delivery person only the orders assigned to them.	Must	Validated
FR-21	The system shall give the delivery person, for each assigned order, the pickup restaurant, the customer's name, phone and workplace, the items, and the cash amount to collect.	Must	Validated
FR-22	The system shall allow the assigned delivery person to mark a delivery as started.	Must	Validated (states Proposed)
FR-23	The system shall allow only the assigned delivery person to confirm a started delivery, recording the confirmation against that order.	Must	Validated
FR-24	The system shall show a customer the current status of their own orders, updated without reloading the page.	Should	Validated

4.7 Payment

ID	Requirement	Priority	Status
FR-25	The system shall allow a customer to choose only among the payment methods the restaurant has enabled.	Must	Validated
FR-26	**(rev.)** When a restaurant enables cash on delivery, the system shall record the payment as pending at ordering and as paid when the delivery person confirms the cash was collected.	Must	Validated
FR-27	**(rev.)** When an approved Tunisian e-payment provider is integrated, the system shall take payment through it at ordering and shall not create the order if the payment is declined.	Should	Validated (provider unresolved)
FR-28	The system shall record each order's payment method and payment status.	Must	Validated

4.8 Planning support

ID	Requirement	Priority	Status
FR-31	**(rev.)** The system shall show authorized restaurant staff a next-service quantity forecast and range per menu item, computed only from aggregated order counts. The forecast shall not create, change, accept or reject any order, payment, eligibility or menu record.	Should	**Proposed** (ADR-0004; stakeholder validation pending)

4.9 Business rules added after the specification

ID	Rule	Status
BR-17	**(rev.)** An order shall be accepted only if both the restaurant and the customer's workplace are located in the service area.	Team decision (ADR-0007)
BR-18	**(new)** A restaurant located outside the service area cannot join FoodGO, even to deliver into it.	Team decision (ADR-0007)

BR-01–BR-16 are unchanged (spec Appendix A5).

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5. Non-functional requirements

Targets are the specification's **Proposed** targets unless marked otherwise. Tool names were moved out of the requirements into the SDD after the peer review.

ID	Category	Requirement	Priority	How it is checked
NFR-01	Performance	**(rev.)** With 300 orders already placed for the day at one restaurant, 95 % of requests shall complete within 500 ms and 99 % within 1.5 s; order submission within 2 s, excluding the payment provider.	Must	`npm run benchmark`
NFR-02	Security	**(rev.)** The system shall authenticate every user, allow each role only its own functions, limit restaurant staff and delivery persons to their own restaurant's data, store credentials only as salted hashes, and slow down repeated failed sign-ins.	Must	Integration tests
NFR-03	Privacy	Customer data shall be visible only to authorized parties (BR-15), and no customer data shall be sent to the LLM.	Must	Scope tests, prompt test
NFR-04	Reliability	The system shall never lose, duplicate or half-save an order.	Must	Transaction and retry tests
NFR-05	Maintainability	**(rev.)** Business rules shall be separated from the user interface and from data storage, and no function shall exceed a cyclomatic complexity of 10.	Should	Lint in CI
NFR-06	Testability	Each feature shall have automated tests, with at least 60 % line coverage of server code.	Must	Coverage in CI
NFR-07	Usability	A customer shall be able to place an order in at most 4 screens, and shall see the ordering deadline before ordering.	Should	UI review
NFR-08	Accessibility	The user interface shall meet WCAG 2.2 level AA.	Should	Audit (planned)

ID	Category	Requirement	Priority	How it is checked
NFR-09	Observability	Every security-relevant action (sign-in, provisioning, order, assignment, delivery, payment) shall be recorded in an audit log, and the service shall expose a health check.	Should	Admin monitor
NFR-10	Interoperability	**(rev.)** Each external provider (payment, SMS/OTP, notification, LLM) shall be replaceable without changing business rules.	Should	Adapter tests
NFR-11	Availability	The system shall be available 99.5 % of each month during the service window. **The service window is still a business decision**, so this target cannot be tested yet.	Should	After deployment
NFR-12	Portability	**(rev.)** The system shall install and run on a fresh machine with a single install command and no external service.	Could	Fresh-clone check
NFR-13	Scalability	**(new, Proposed)** The system shall handle the morning ordering peak of the whole service area. Target numbers (restaurants, customers, orders per peak) are set after the Sprint 2 interviews. Serving another area requires a new ADR.	Should	Benchmark at the agreed numbers

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6. Use cases and models *(placeholder)*

Use-case diagram and UC-01 / UC-04 descriptions: Lab 3 worksheet, Step 4. Full UC register: Lab 4 SDD, Appendix A. UML: `docs/uml/`.

7. Appendices *(placeholder)*

Traceability matrix (`docs/requirements/TRACEABILITY.md`), unresolved items (spec Appendix A6, minus items 35 and part of 10), glossary (Section 1).